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%sheikh wasimuddin lab 10 CSEIOT61
clc;
clear;
close all;
rng(21);
nodes=70;
area=110;
range=30;
initialEnergy=1;
iterations=110;
px=rand(1,nodes)*area;
py=rand(1,nodes)*area;
E=initialEnergy*ones(1,nodes);
kind=randi([1 4],1,nodes);
temp=22+6*rand(1,nodes)+3*sin(py/20);
humid=50+25*rand(1,nodes)-8*cos(px/18);
air=35+35*rand(1,nodes)+15*exp(-((px-75).^2+(py-30).^2)/700);
soil=30+45*rand(1,nodes)+10*exp(-((px-30).^2+(py-75).^2)/900);
BS=[50 105];
energyHistory=zeros(1,iterations);
coverageHistory=zeros(1,iterations);
deliveryHistory=zeros(1,iterations);
activeHistory=zeros(1,iterations);

figure('Color','w','Position',[120 100 850 650]);
hold on;
gscatter(px,py,kind);
plot(BS(1),BS(2),'kp','MarkerSize',14,'MarkerFaceColor','k');
xlim([0 area]);
ylim([0 110]);
grid on;
box on;
xlabel('X Coordinate (m)');
ylabel('Y Coordinate (m)');
title('Heterogeneous Wireless Sensor Network');
legend('Temperature','Humidity','Air Quality','Soil Moisture','Base
Station','Location','southoutside');

for t=1:iterations
alive=E>0;
sent=0;
received=0;
for n=1:nodes
if ~alive(n)
continue;
end
ds=hypot(px(n)-BS(1),py(n)-BS(2));
if ds<=range
d=ds;
relay=0;
else
others=find(alive);
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others(others==n)=[];
if isempty(others)
continue;
end
dlist=hypot(px(others)-px(n),py(others)-py(n));
nearby=others(dlist<=range);
if isempty(nearby)
continue;
end
toBS=hypot(px(nearby)-BS(1),py(nearby)-BS(2));
[~,q]=min(toBS);
relay=nearby(q);
d=hypot(px(n)-px(relay),py(n)-py(relay));
end
Es=0.00012;
Et=0.00035+0.000012*d^2;
E(n)=max(0,E(n)-Es-Et);
sent=sent+1;
if relay==0
received=received+1;
elseif E(relay)>0
E(relay)=max(0,E(relay)-0.00018);
received=received+1;
end
end
active=find(E>0);
cells=8;
count=0;
for a=1:cells
for b=1:cells
cx=(a-0.5)*area/cells;
cy=(b-0.5)*area/cells;
if ~isempty(active)
dist=hypot(px(active)-cx,py(active)-cy);
if min(dist)<=range
count=count+1;
end
end
end
end
coverageHistory(t)=100*count/(cells*cells);
if sent>0
deliveryHistory(t)=100*received/sent;
end
energyHistory(t)=sum(E);
activeHistory(t)=nnz(E>0);
end

figure('Color','w','Position',[120 100 900 650]);
subplot(2,2,1);
plot(energyHistory,'LineWidth',1.7);
grid on;
box on;
xlabel('Round');

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ylabel('Energy (J)');
title('Remaining Energy');

subplot(2,2,2);
plot(coverageHistory, 'LineWidth',1.7);
grid on;
box on;
xlabel('Round');
ylabel('Coverage (%)');
title('Network Coverage');
ylim([0 105]);

subplot(2,2,3);
plot(deliveryHistory, 'LineWidth',1.7);
grid on;
box on;
xlabel('Round');
ylabel('Delivery (%)');
title('Data Reliability');
ylim([0 105]);

subplot(2,2,4);
plot(activeHistory, 'LineWidth',1.7);
grid on;
box on;
xlabel('Round');
ylabel('Active Nodes');
title('Network Lifetime');

sgtitle('WSN Performance Evaluation');

figure('Color','w','Position',[120 100 900 650]);
subplot(2,2,1);
scatter(px(kind==1),py(kind==1),55,temp(kind==1),'filled');
colorbar;
grid on;
box on;
xlabel('X Coordinate (m)');
ylabel('Y Coordinate (m)');
title('Temperature');
xlim([0 area]);
ylim([0 area]);

subplot(2,2,2);
scatter(px(kind==2),py(kind==2),55,humid(kind==2),'filled');
colorbar;
grid on;
box on;
xlabel('X Coordinate (m)');
ylabel('Y Coordinate (m)');
title('Humidity');
xlim([0 area]);
ylim([0 area]);

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subplot(2,2,3);
scatter(px(kind==3),py(kind==3),55,air(kind==3),'filled');
colorbar;
grid on;
box on;
xlabel('X Coordinate (m)');
ylabel('Y Coordinate (m)');
title('Air Quality');
xlim([0 area]);
ylim([0 area]);

subplot(2,2,4);
scatter(px(kind==4),py(kind==4),55,soil(kind==4),'filled');
colorbar;
grid on;
box on;
xlabel('X Coordinate (m)');
ylabel('Y Coordinate (m)');
title('Soil Moisture');
xlim([0 area]);
ylim([0 area]);

sgtitle('Environmental Data Analysis');

fprintf('\nWIRELESS SENSOR NETWORK RESULTS\n');
fprintf('=====\n');
fprintf('Sensor Nodes          : %d\n',nodes);
fprintf('Monitoring Area        : %d x %d m\n',area,area);
fprintf('Communication Range     : %.1f m\n',range);
fprintf('Simulation Rounds       : %d\n',iterations);
fprintf('Active Nodes           : %d\n',activeHistory(end));
fprintf('Remaining Energy        : %.4f J\n',energyHistory(end));
fprintf('Mean Coverage           : %.2f %%\n',mean(coverageHistory));
fprintf('Mean Reliability        : %.2f %%\n',mean(deliveryHistory));
fprintf('Final Coverage          : %.2f %%\n',coverageHistory(end));
fprintf('Final Reliability       : %.2f %%\n',deliveryHistory(end));
fprintf('=====\n');

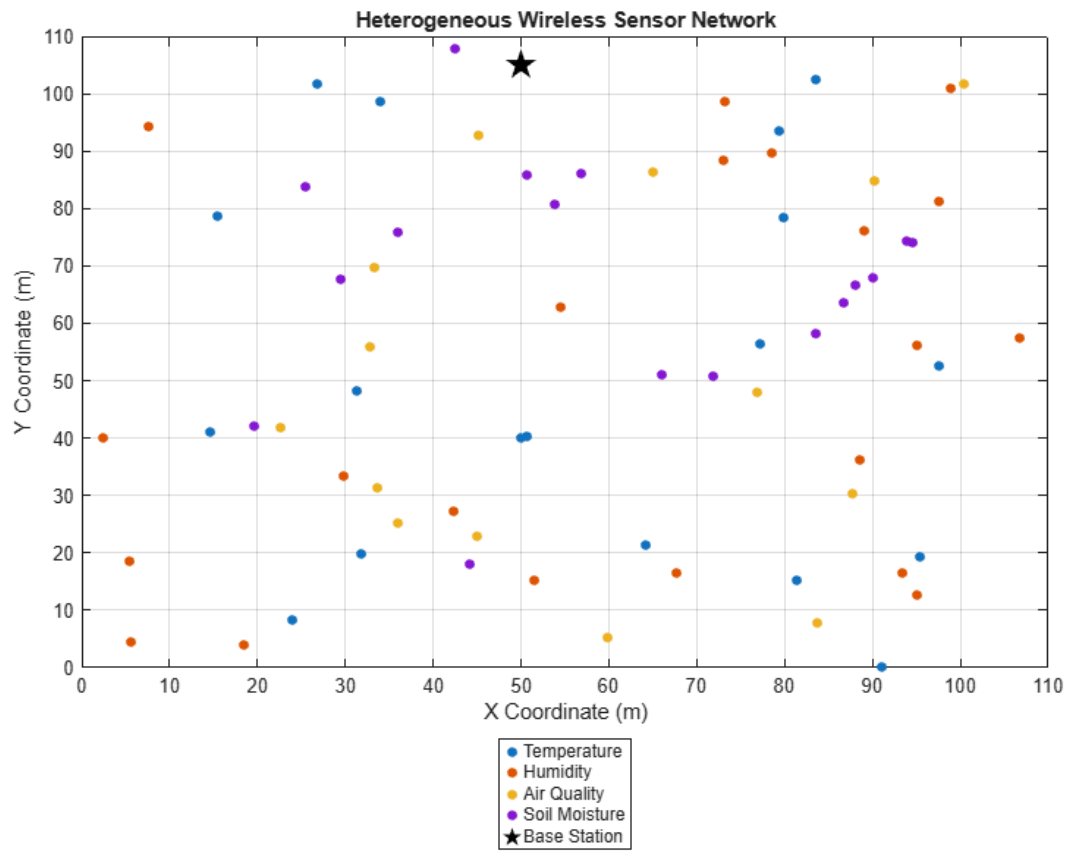
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WIRELESS SENSOR NETWORK RESULTS

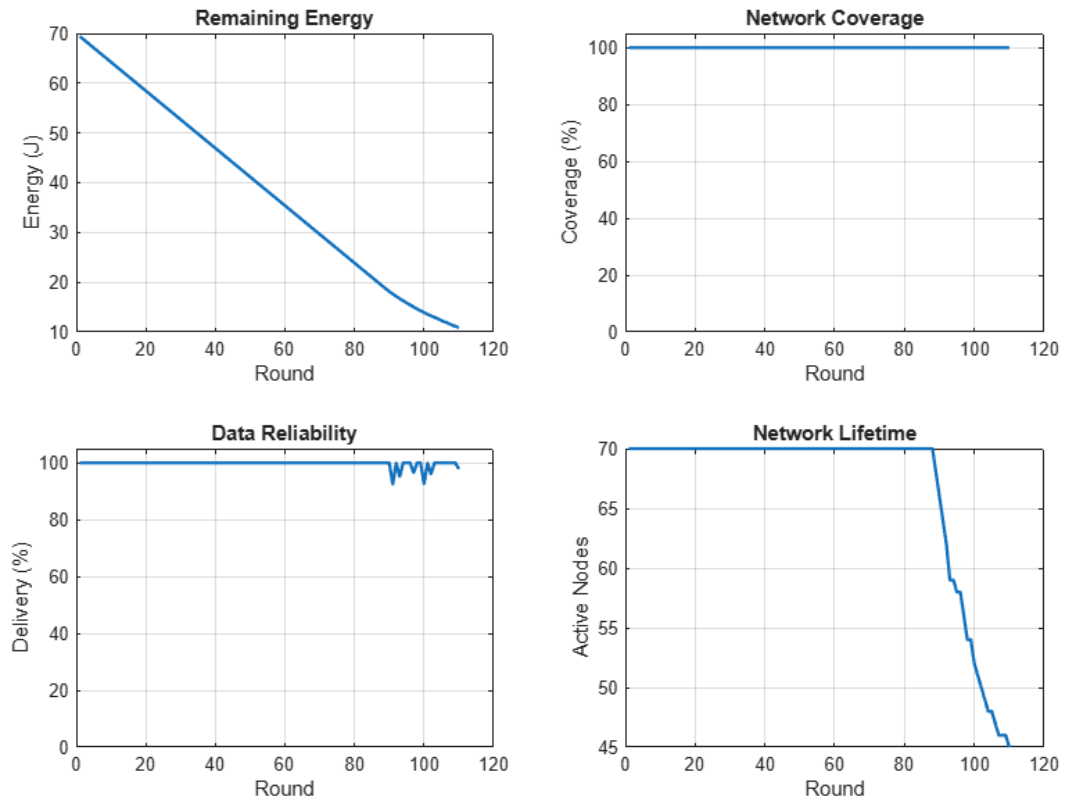
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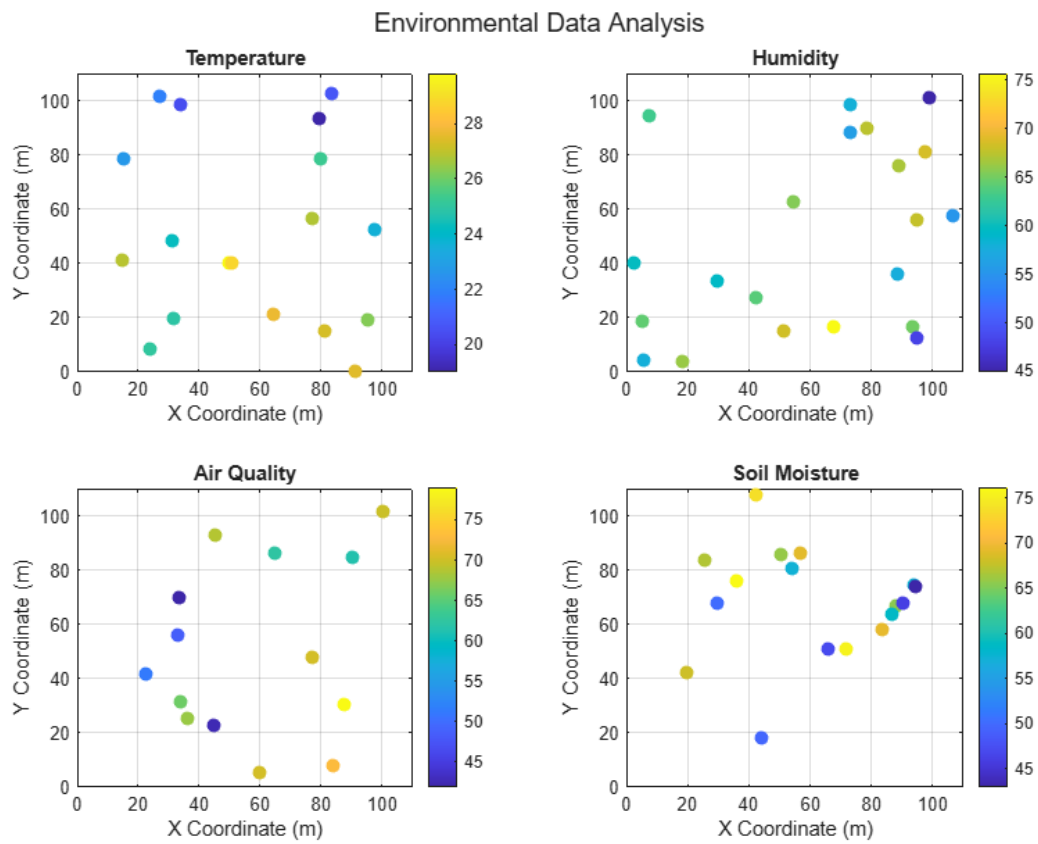
=====
Sensor Nodes          : 70
Monitoring Area       : 110 x 110 m
Communication Range    : 30.0 m
Simulation Rounds      : 110
Active Nodes          : 45
Remaining Energy       : 10.8531 J
Mean Coverage          : 100.00 %
Mean Reliability       : 99.73 %
Final Coverage         : 100.00 %
Final Reliability      : 97.83 %
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WSN Performance Evaluation





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